

Alen Gigi Joseph

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Summary

Robotics Engineer specializing in perception, localization, and autonomous systems. My work combines computer vision, geometric estimation, and robotics software to solve real-world measurement, navigation, and spatial understanding challenges.

Experience

- SkyMul | Robotics Engineer** **December 2025 – Present**
- Developed camera calibration and geometric estimation pipelines for accurate spatial measurements in construction robotics applications.
 - Implemented multi-view pose estimation and coordinate-frame alignment algorithms to determine camera and object poses in real-world environments.
 - Engineered vision-based measurement workflows integrating image observations with external surveying systems through spatial registration and coordinate transformations.
 - Established validation, benchmarking, and reprojection error analysis pipelines to improve localization accuracy and geometric consistency.
- FISAT Rover Team | Project Guide and Mentor** **September 2025 – November 2025**
- Established and structured a 40+ students interdisciplinary team for the International Rover Challenge.
 - Directed training, task allocation, and project execution to deliver a competition-ready rover in under 10 weeks
- AutoInfer (Client: Ati Motors) | Robotics Engineer Intern** **May 2025 – August 2025**
- Developed monocular camera-based pose estimation systems for industrial trolleys used alongside autonomous robots.
 - Worked on pedestrian and obstacle detection pipelines for collision avoidance.
 - Assisted in system validation, debugging, and performance tuning of perception modules in Linux environments.
- GE Aerospace | Project Intern** **January 2025 – March 2025**
- Contributed to automation and validation activities in a multi-modal factory environment.
 - Supported debugging and verification of integrated software-hardware workflows.
 - Collaborated with cross-functional teams to diagnose and resolve system-level issues

Projects

- A Hybrid Quadruped-Wheeled Robot**
- Engineered a terrain-adaptive hybrid locomotion platform combining wheeled mobility with quadrupedal traversal, reducing system cost by over 70% compared to comparable robotic platforms.
 - Developed locomotion control and gait-transition mechanisms enabling sub-250 ms mode switching, 200 Hz real-time control, and obstacle traversal of up to 13 cm across uneven terrain.
 - Architected a distributed control framework integrating embedded motor control, sensor feedback, and high-level autonomy modules for coordinated multi-modal locomotion.
 - Published the work at the 9th International Conference on Robotics and Automation Sciences (ICRAS 2025), Osaka, Japan.
- Low-Cost Autonomous Mobile Robot (Mini Project)**
- Engineered a differential-drive autonomous mobile robot integrating LiDAR, IMU, motor control, onboard computing, and autonomy software for indoor navigation.
 - Implemented localization, mapping, and navigation pipelines using probabilistic state estimation, graph-based optimization, path planning, and trajectory tracking algorithms.

- Diagnosed and mitigated localization drift, sensor synchronization errors, and navigation failures through system-level testing and calibration.

TripSage: An optimised route planner app:

- Developed a route optimization platform integrating traffic analytics, map services, and real-time congestion estimation.
- Built backend services for camera-based vehicle speed estimation and traffic flow analysis.
- Integrated geospatial data pipelines using OpenStreetMap APIs.

ROS Android App (Open Source)

- Developed an Android application for real-time robot monitoring, control, and configuration integrated with ROS2 middleware.
- Implemented real-time telemetry visualization and bidirectional robot communication using ROS topics and WebSocket-based data streaming.

Urban BloomPord (Rural Investors Meet, Kerala, 2023)

- Built an IoT system to monitor environmental parameters and automate plant care.
- Implemented relay-based control on Raspberry Pi and a mobile app for remote operation.

Publications

- Development of a Terrain Adaptive Quadruped Wheeled Robot, 9th International Conference on Robotics and Automation Sciences (ICRAS 2025, Osaka, Japan)
- Development of a Low-Cost Autonomous Mobile Robot
- Cost Effective Indoor Localization using BLE beacons and LoRa WAN Gateways

Education

Federal Institute of Science And Technology – Kerala, India
Bachelor of Technology in Electronics and Instrumentation (3.04/4.0 GPA)

Graduated May 2025

Achievements and Extra-Curricular

- **Memberships:** International Society of Automation (ISA), FAB Lab (Technical Lead), IEEE
- Lead and Resource person, Robotics Lab, EIE-FISAT
- Eyantra Robotics Competition 2025, IIT Bombay (Top 20 National out of 600+ teams)

Skills/Courses

- Foundations: Linear algebra, optimization, robot kinematics, Projective geometry, multi-view geometry, machine learning, deep learning
- Frameworks & Tools: Linux, ROS, Git, Docker, ESP-IDF, PyTorch, Android
- Programming: C, C++, Kotlin; scripting in Python and Bash
- Microcontrollers, Sensor Integration, Communication Interfaces: CAN, SPI, I2C, UART
- Simulation & Design: MATLAB, Simulink, Isaac Sim, Gazebo, Proteus, Fusion 360